

# Uzair Akbar

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## EDUCATION

**M.Sc. Computer Science** @ *Georgia Institute of Technology*

**GPA:** 3.78 / 4.0 | **Dec. 2025**

**M.Sc. Communications Eng.** @ *Technical University of Munich*, Germany

**GPA:** 3.65 / 4.0 | **June 2021**

**B.Sc. Electrical Eng.** @ *National University of Sciences & Technology*, Pakistan

**GPA:** 3.59 / 4.0 | **June 2017**

## SELECTED PUBLICATIONS ([Google Scholar](#))

**U. Akbar et al.** “Symmetry-Informed Causal Partial Identification.” *arXiv:2610.09230*, 2026. **(preprint)**  

**U. Akbar et al.** “An Analysis of Causal Effect Estimation using Outcome Invariant Data Augmentation.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. **(spotlight; top 3%)**   

A. Savkin, T. Lapotre, K. Strauss, **U. Akbar** & F. Tombari, “Adversarial Appearance Learning in Augmented Cityscapes for Pedestrian Recognition in Autonomous Driving.” *ICRA*, 2020. 

## PROFESSIONAL EXPERIENCE ([LinkedIn](#))

**Software Engineer Intern** @ *Hewlett Packard Enterprise*, USA

**June '24—Aug. '24**

- Built CI/CD pipeline (`python` + `jenkins` + `JFrog`) to test/build/release/rollback VM templates for QA workloads.
- Integrated pipeline with dynamic KVM provisioning platform (`k8s` + `KubeVirt`) for scalable env. management.

**Software Engineer** @ *OMMAX*, Germany

**Nov. '22—June '23**

- Designed end-to-end Salesforce ETL pipeline (`Airflow` + `PostgreSQL`), improving data throughput by 58% (by refactoring `python` → `postgres`) and sales lead identification by 63% (by reducing data redundancy & duplicates).
- Built a serverless `FastAPI` microservice on AWS Lambda (`docker`) for `XGBoost`-based ticket classification and RegEx error code extraction. Served endpoints to a CRM platform to automate ticket routing and categorization.

**Data Scientist** @ *Auto1 Group*, Germany

**June '22—Oct. '22**

- Engineered `LightGBM` predictor for car refurbishment costs; €160 MAE decrease (~27.5% improvement) vs. prod.
- Automated training and eval. workflows; data extraction/transformation (Redshift) & model checkpointing (S3).

**ML Research Intern** @ *Max Planck Institute of Intelligent Systems*, Germany

**Apr. '20—Feb. '21**

- Developed a method to reduce hidden confounding bias w/ data-augmentation + instrumental variable regression.
- Foundations of the later [spotlight](#) @ NeurIPS 2025, and ongoing work on [symmetry-informed causal inference](#).

## TEACHING EXPERIENCE

**Teaching Assistant** (Reinforcement Learning [CS8803](#), [CS7642](#)) @ *Georgia Tech*, USA

**Aug. '24—Present**

- Authored RL projects for ~ 300 students/semester, including a `python` package for local [DeepRacer](#) deployment.
- Built a containerized simulation service with multi-runtime support (`docker` (`compose`) + `podman` + `apptainer`).
- Developed a client-side `gymnasium` wrapper to interface with the simulation service using `ZeroMQ` sockets. 

**Teaching Assistant** (Approx. Dynamic Programming [EI7649](#)) @ *TU Munich*, Germany

**Oct. '19—Mar. '20**

- Held weekly tutorials discussing problem-set solutions, graded assignments, and mentored student group projects.

## TECH. STACK ([GitHub](#))

**Languages:** `Python`, `C++`, `SQL`, `Bash`

**Data & Backend:** `PostgreSQL`, `Airflow`, `Redis`, `FastAPI`

**Testing & CI/CD:** `Jenkins`, `JFrog`, `Git`

**Systems & Orchestration:** `Docker`, `Linux`, `RabbitMQ`, `K8s`

**Cloud:** `AWS` (EC2, EKS, Lambda, S3, RDS, Redshift)

**ML:** `PyTorch`, `TensorFlow`, `sk-learn`, `Gymnasium`

## REVIEWING

NeurIPS 2026